EQUIPMENT NAME:	**FIM III Booster Pump Variable Speed Drives**
EQUIPMENT NUMBER:	**F23-PP-651-VS** **F23-PP-652-VS**

Procurement Package			"EL006" - HV Variable Speed Drives			
Equipment No.	**Voltage Ratio (kV)**	**Motor Rating (kW)**	**VSD Duty (150% for 60s)**	**Motor FLA (A)**	**Cooling Method**	**Equipment Name**
F23-PP-651-VS	11/3.3	800	Heavy Duty	171.12	Air	FIM III Booster Pump 1 Variable Speed Drive
F23-PP-652-VS	11/3.3	800	Heavy Duty	171.12	Air	FIM III Booster Pump 2 Variable Speed Drive
Reference Documents	2000-F23-SLD-EL-40001 – FIM III Booster Station F23-SW-601 HV Switchboard Single Line Diagram 2000-F23-SLD-EL-40002 – FIM III Booster Station F23-SW-601 HV Switchboard Key Interlocking Single Line Diagram					
Reference Specifications	2000-F00-STS-GE-10002 - Site Specific Information 2000-F00-STS-GE-10001 - Packing and Shipping Specification SPEC-00-G-05 – Protective Coating 2000-F00-STS-EL-10004 - High Voltage Electrical Testing Specification					

Scope of Supply
Work Included • Design, and supply of HV Variable Speed Drives including, but not limited to: ○ HV Variable Speed Drives   ○ Accessories. As required to meet specification • Vendor Documentation, certificates, drawings and data. Vendor to refer to Vendor Drawing and Document Requirements List below. • All ancillary items and special tools normally supplied with, or, required to adjust and maintain all components associated with the equipment. • Packaging suitable for sea and road transport, and delivery to the delivery point.
Work Excluded • Spare parts • Offloading of equipment at site • Installation and commissioning at site

Description	**Unit**	**Duty**	**Supplier**	**Notes**	**Rev**
General					
VSD Type		11/3.3kV, Air cooled	Complies	3	0
Make		Siemens	Complies	3	0
Model		SINAMICS Perfect Harmony GH180 GenVe-11-3.3-200-1250 6SR5502-0NG41-2BH0-Z A60+B43+C68+D02+D15+D76+G28+G47+K31+K73+K86+L03+L55+M36+M97+N27+N94+T03+T76+U58+V04+W68+X30+X32+N37+K68	6SR55020NF412BH0-Z We offer 932kW, 260A drive to meet the requested overload of 150% (171.12A). The offered VSD overload capability is 166% (1/10 min).	3	0
Country of Manufacture				2	0
Compliance with Australian Standards	Comply /Do Not Comply	Comply with AS61800 (IEC61800), vendor to confirm	The VSDs are built to the specific product standards AS/IEC 61800.5.1 and 61800.3.	3	0
Environmental Conditions					
Plant Location		Western Australia		1	0
Service Location		Indoor, within a switchroom in a temperature controlled, air-conditioned environment	Complies	1	0
Site Data		Refer to 2000-F00-STS-GE-10002 – Site Specific Information		1	0
VSD Supply Voltage					
Supply		3P+E	Complies	1	0
Input Voltage	kV	11 ± 10%	Complies	1	0
Frequency	Hz	50 ± 2%	Complies	1	0
Neutral Earthing		NER – at Upstream Transformer	Complies	1	0
General					
Rated Line Voltage (Input)	kV	11	Complies	1,4	0

Description	Unit	Duty	Supplier	Notes	Rev
Rated Output Voltage	kV	3.3	Complies	1	0
Line Side Cable		3C 95mm² Cu 6.35/11kV XLPE/CWS/PVC/NY/PVC	Complies	1	0
Load Side Cable		3C 185mm² Cu 3.8/6.6kV XLPE/CWS/PVC/NY/PVC	Complies	1	0
Control Cable		1.5mm² core and earth unscreened for digital, 1.5mm² screened twisted pair for analogue	Complies	1	0
VSD Front End Components					
Rectifier Type		2Q		3	0
Type of Thyristor		Diodes	Complies	3	0
VSD Inverter Components					
PWM Power Modules	Yes/No	Yes		1	0
Type of Thyristor		Diodes	Complies	3	0
Peak Inverse Voltage	kV	1.7	Complies	3	0
VSD Performance					
1 min Cyclic Overload Period		1 minute every 10 minutes		3	0
Normal Duty Cyclic Overload Capacity	%	≥ 110%	Please refer to the provided VSD data sheet, as it depends on the specific application	3	0
Heavy Duty Cyclic Overload Capacity	%	≥ 150%	Please refer to the provided VSD data sheet, as it depends on the specific application	3	0
10 sec Peak Overload Period		1 minute every 10 minutes		3	0
Normal Duty Peak Overload Capacity	%	≥ 150%	Please refer to the provided VSD data sheet, as it depends on the specific application	3	0
Heavy Duty Peak Overload Capacity	%	≥ 180%	Please refer to the provided VSD data sheet, as it depends on the specific application	3	0
Starting Torque Requirement (% Full Load motor torque)	%		Please refer to the provided VSD data sheet, as it depends on the specific application	2	0
VSD Induced Torque Pulsations	%	<1	Complies	1	0
THD(I) at 100% Full Load	%	<5	Complies	1	0
dV/dt	kV/s	<3000V/µs	Complies	3	0
Efficiency at 100% Full Load (at rated speed)	%	96.5	Complies	3	0
Efficiency at 75% Full Load	%	96.4	Complies	3	0
Max Heat Dissipation @ 100% Full Load	kW	3.5% of the motor rating (kW)	Complies	3	0
Cooling Method		Air Cooled	Complies	1	0
Power Disturbance Ride-through (loss of cycles immunity, voltage sag immunity)		Min 5 cycles (MV ride-through)	Complies	3	0
VSD Protections					
Coolant	Yes/No	Yes	Complies	3	0
Fan Failure	Yes/No	Yes	Complies	3	0
Thyristor/IGBT/IGCT Temperature Monitoring	Yes/No	Yes	Complies	3	0
VSD Safety Devices					
Interlocking for HV Power section panels/doors (Fortress Keys) – if required	Yes/No	Yes, Fortress key, refer to 2000-F23-SLD-EL-40001	The G5 GH180 model features liftoff bolted covers that require tools for opening. In line with Owner site standards and the configuration of previously delivered drives, we have offered a Fortress Interlock exclusively on the auxiliary cabinet door. Additionally, we have provided a pad-lockable handle on the G5 control cabinet door, aligning with the previous drive configurations delivered to the Owner site.	3	0
Timer to prevent opening HV power section panels/doors until DC capacitors have discharged	Yes/No	Yes	Referring to the Owner site standard, we did not include the Time Delay Unit (TDU) in our offer. However, Innomotics can provide the TDU as a standalone unit for installation by the customer on-site, if required.	3	0
Earthing of Metal work	Yes/No	Yes	Complies	1	0
Emergency Stop on Front Panel	Yes/No	Yes	Complies	3	0
Safety Relay Installed	Yes/No	Yes, comply with Safe Torque Off (Option K86)	Complies	3	0

Description	Unit	Duty	Supplier	Notes	Rev
Prevention of Unexpected Start-up	Yes/No	Yes	Complies	3	0
Safe Torque-Off	Comply /Do Not Comply	Safe Torque Off function shall comply with: • Machines Category 3 • Level (SIL) 2 • Machinery – Stopping Category 0	Complies	1	0
Process Stop	Yes/No	Yes	Complies	3	0
Run Enable	Yes/No	Yes	Complies	3	0
Arc Flash Detection Relay and Sensors	Yes/No	Yes	Complies	3	0
Motor Protection					
Integrated Motor Protection Relay	Yes/No	Yes	Complies	3	0
Motor PT100 (RTD) Relay	Yes/No	Yes (Option A60)	Complies	3	0
VSD Enclosure					
Arc Resistant Design and associated protection as per IEC 62477-2. Vendor to state what protection equipment is required upstream of VSD to achieve an arc-resistant design.			N/A. Innomotics can supply the Arc resistance cabinet under the M57 option if required on site	2	0
Degree of Ingress Protection		IP42	Complies	3	0
Removable Filters from Outside	Yes/No	Yes	Complies	3	0
Cabinets Steel Gauge	mm	2.5	Complies	3	0
Door Latches		Fortress Locks	Bolted Cover	3	0
Access Required		Front and Top only	Complies	1	0
Cable Entry		Bottom Entry	Complies	1	0
Gland Plate		5mm Al Brass	Complies	3	0
Switchroom Minimum Ceiling Height for Maintenance (removal of fans etc.)		300mm above VSD's height	Complies	3,5	0
VSD Controls & Indications					
LCD Display	Yes/No	Yes	Complies	1	0
Speed/Torque/Droop Control Mode		Closed Loop Vector Control (0.1% Dynamic Speed Error) or Open-loop Vector Control (0.5%). Vendor to Advise	N/A for pump applications	3	0
Switching Frequency		Switching frequency at full load: > 3 kHz	Complies	3	0
Acceleration Time Range (min-max)	s	0-3200	Complies	3	0
Starting curves available		Programmable S-Curve, Linear	Complies	3	0
Motor Speed Feedback encoder card			N/A for pump applications	2	0
Output Frequency Range	Hz	0-75	Complies	3	0
Output Speed Limits	%	200	Complies	3	0
No. of on-board VSD settings/configuration files		3	Complies	3	0
Communications		Modbus TCP	Complies	1	0
Digital Inputs		20 more as option POA	Complies	3	0
Digital Outputs		16 more as option POA	Complies	3	0
Analog Inputs		3 more as option POA	Complies	3	0
Analog Outputs		2 more as option POA	Complies	3	0
Control Voltage	V	240VAC UPS Supply	K68 option offered / Control voltage 240V AC by customer.	3	0
Control Supply Load	VA/W		TBC	2	0
Auxiliary Supply Cable		3C+N+E	Complies	1	0
Auxiliary Supply Voltage (Cooling fans, motor heater)	V	415V AC, 3ph, 50Hz	Complies	1	0
Auxiliary Supply Voltage from on-board Tx	Yes/No		K68 option offered / Control voltage 240V AC by customer.	2	0
Auxiliary Supply Load	VA/W		Tbc	2	0
Shipping and Delivery Information					
Dimensions	mm	Length x Width x Height	2905 x 1143 x 2779	2	0
Weight	kg		5000	2	0

Notes
1. Fixed. Vendor shall comply with this data
2. Vendor to Advise
3. Preferred, Vendor to advise and provide reasoning for difference
4. Vendor shall ensure VSD is suitable of operating within the stated input voltage range
5. Vendor to specify minimum indoor ceiling height of switchroom(s) that shall house their HV VSDs. If the minimum ceiling height differs for each VSD frame / kW size, the Supplier shall specify the minimum indoor ceiling height for each VSD frame / kW size.

Vendor Drawing and Document Requirements List (VDDR)

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
A		**Production Schedule**				
	A01	Production Schedule / Bar Chart		1E	6-8	
	A02	Monthly Report		Monthly		
B		**Registers and General Drawing Requirements**				
	B01	Drawing Schedule / Register (Drawings Expected)		1E	6-8	
	B02	General Arrangement with Foundation Loads (in AutoCAD 2010 and PDF)		1E	6-8	1E
	B03	Cross-section		1E	6-8	1E
C		**Mounting and Erection Drawings**				
	C01	Foundation and Mounting Details (Provided in B02)		1E	6-8	1E
	C02	Pipe Spools				
	C03	Outline Dimensions, Cut-out and Support Details (Provided in B02)		1E	6-8	1E
	C04	Match Marked Erection Drawings				
	C05	Motor GA (with Fixing and Shaft Details, and Overall Dimensions)				
D		**Electrical Drawings (in electronic form as appropriate)**				
	D01	Single Line Diagrams		1E	6-8	1E
	D02	Schematic Drawings		1E	6-8	1E
	D03	Terminations / Loop Diagrams		1E	6-8	1E
	D04	Panel Layout and Fabrication Details		1E	6-8	1E
	D05	Nameplate		1E	6-8	1E
	D06	Cable Block Drawings				
	D07	Electrical and Instrumentation GAs		1E	6-8	1E
E		**Process and Piping Drawings**				
	E01	Process Flow Sheets				
	E02	Process and Instrument Diagram				
	E03	Pneumatic or Hydraulic Diagram				
	E04	Lubrication Diagram				
F		**Technical Data**				
	F01	Equipment Information Schedule				
	F02	Electrical Performance data sheet		1E	6-8	
	F03	Equipment List		1E	6-8	1E
	F04	Vendor's Surface Protection Details				
	F05	Lubrication Data Schedule				
	F06	Material Schedule				
	F07	Bolt Schedule				
	F08	Fan Performance Curves				
	F09	Pump Performance Curves (with Duty Points Marked)				
	F10	Cable Schedule				
	F11	Vendor Catalogue Information				
	F12	Calculations				
	F13	Completed $_{\text{Engineer}}$ Datasheets	1E			
	F14	Motor Datasheets	1E			
	F15	Electrical Equipment List				
	F16	Instrument List				
	F17	I/O List		1E	6-8	
	F18	Communications Addressing and Signal Mapping List		1E	6-8	

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
G		**Inspection, Certification and MDR**				
	G01	Inspection and Test Plans		1E	6-8	
	G02	Inspection and Test Procedures (generic)		1E	6-8	
	G03	Quality Plans and Procedures (generic)		1E	6-8	
	G04	Manufacturer's Data Report (MDR) – Index / Contents		1E	6-8	
	G05	Manufacturer's Data Report (MDR)		1E	6-8	
	G06	Material Certificates				
	G07	Statutory Authority Certificates				
	G08	Test Data and Other Certificates		1E	6-8	
	G09	Welding Procedure / Specification				
	G10	Non-Destructive Testing (NDT) Procedures				
	G11	Material Safety Data Sheets				
	G12	Commissioning Plan and Procedure		1E	6-8	
	G13	Structural Design Certification				
	G14	Performance Guarantee Statement				
H		**Shipping and Packing Details**				
	H01	Material / Shipping List	1E	1E	Delivery -4	
	H02	Package Type and Quantity (Carton, Case, Crate)		1E	Delivery -4	
	H03	Shipping Packages – Weights and Dimensions		1E	Delivery -4	
	H04	Shipping and Handling Procedure		1E	12	
J		**Manuals**				
	J01	Installation, Operation and Maintenance Manual (Preliminary)		1E	Delivery -4	
	J02	Installation, Operation and Maintenance Manual (Final)		1E	Delivery	
K		**Spare Parts Information**				
	K01	Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals)	1E	1E	8	
	K02	Recommended Spare Parts List with Pricing (Not Included in Manuals)	1E	1E	Prelim 4 Delivery +1	
	K03	Special Tools list	1E	1E	8	
M		**General**				
	M01	Control Philosophy and Logic Diagram				
	M02	Functional Description				